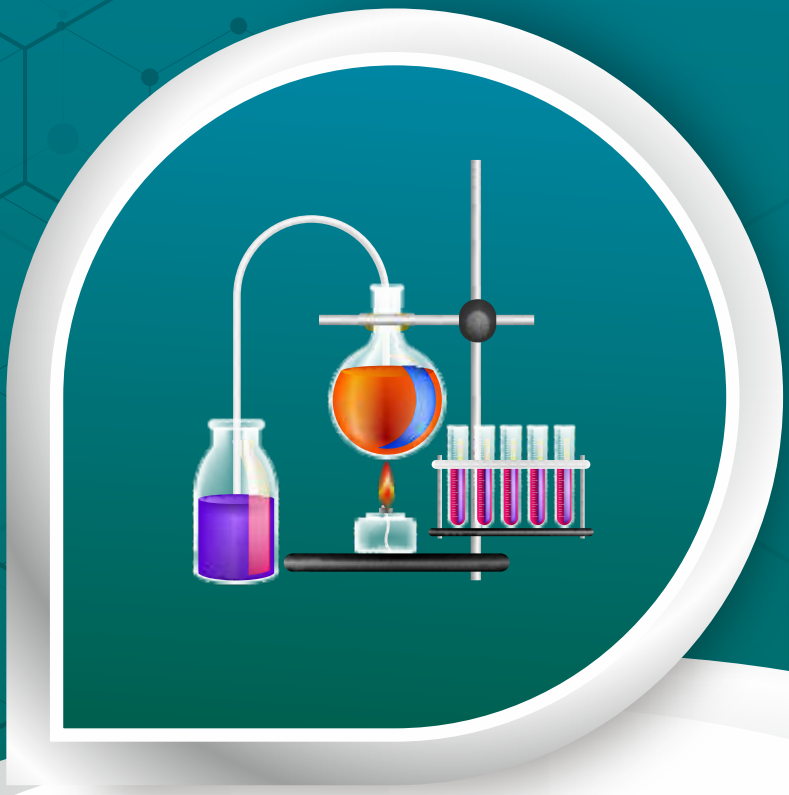
ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success

ORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)



Exercise

Purification and Characteristics of Organic Compounds

(English Medium)

EXERCISE # O-I

1. For the Separation of two immiscible liquids which method (or apparatus) is used?

(1) Chromatography (2) Fractionating column

(3) Fractional distillation (4) Separating funnel **PO0001**

2. A mixture of benzene and chloroform is separated by

(1) Sublimation (2) Separating funnel (3) Crystallization (4) Distillation **PO0002**

3. Which of the following has molecular weight of 92 ?

(1) Toluene (2)Benzene (3)Methylene (4)Propene **PO0003**

4. Aniline is usually purified by

(1) Chromatographic technique (2) Steam distillation

(3) By addition of oxalic acid (4) Fractional crystallization **PO0004**

5. The most suitable method for separation of 1:1 mixture of ortho- and para- nitrophenols is

(1) Sublimation (2) Chromatography (3) Crystallization (4) Steam distillation **PO0005**

6. Distillation under reduced pressure is employed for

(1) Benzene (2) Petrol

(3) Glycerol (4) Organic compounds used in medicine **PO0006**

7. Impure glycerine is purified by

(1) Steam distillation (2) Simple distillation

(3) Vacuum distillation (4) None of the above **PO0007**

8. The latest technique used for purification for organic compuounds in

(1) Chromatography (2) Vacuum distillation

(3) Fractional distillation (4) Crystallisation **PO0009**

9. In paper chromatography

(1) Moving phase is liquid and stationary phase is solid

(2) Moving phase is liquid and stationary phase is liquid

(3) Moving phase is solid and stationary phase is solid

(4) Moving phase is solid and stationary phase is liquid **PO0010**

10. A is a lighter phenol and B is an aromatic carboxylic acid.Separation of mixtue A and B can be carried out easily by using a solution of

(1) Sodium hydroxide (2) Sodium sulphate (3) Calcium chloride (4) Sodium bicarbonate **PO0011**

11. Water and Aniline can be separated in purified form by :

(1) Distillation (2) Fractional distillation

(3) Distillation under reduced pressure (4) Steam distillation **PO0070**

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12.

	List-I (Process of purification)		List-II (Principle involved the process)
(A)	Crystallization	(1)	Liquids which are immiscible in water possessing high boiling point, steam volatile.
(B)	Sublimation	(2)	The compound should be soluble in the solvent at its boiling temperature
(C)	Fractional distillation	(3)	The compound should have high vapour pressure below its melting point
(D)	Steam distillation	(4)	Liquids which have boiling point difference less than 20°C

The correct match is :

(1) A → 2 ; B → 3 ; C → 4; D → 1 (2) A → 2 ; B → 3 ; C → 1; D → 4

(3) A → 4 ; B → 2 ; C → 1; D → 3 (4) A → 3 ; B → 2 ; C → 4; D → 1

PO0071

13. ***Assertion :*** Moving phase is liquid and stationary phase is solid in paper chromatography.
Reason : Paper chromatography is used for analysis of polar organic compounds.

(1) If both ***Assertion*** and ***Reason*** are true and the ***Reason*** is the correct explanation of the ***Assertion*** .

(2) If both ***Assertion*** and ***Reason*** are true but ***Reason*** is not the correct explanation of the ***Assertion*** .

(3) If ***Assertion*** is true but ***Reason*** is false.

(4) If ***Assertion*** is false but ***Reason*** is true.

PO0072

14. ***Assertion :*** Thiophene present in commercial benzene as an impurity can be removed by shaking the mixture with cold concentrated H_2SO_4.
Reason : Thipohene is a heterocyclic aromatic compound.

(1) If both ***Assertion*** and ***Reason*** are true and the ***Reason*** is the correct explanation of the ***Assertion*** .

(2) If both ***Assertion*** and ***Reason*** are true but ***Reason*** is not the correct explanation of the ***Assertion*** .

(3) If ***Assertion*** is true but ***Reason*** is false.

(4) If ***Assertion*** is false but ***Reason*** is true.

PO0073

15. ***Assertion :*** Refining of petroleum involves fractional distillation.
Reason : Fractional distillation involves repeated distillation.

(1) If both ***Assertion*** and ***Reason*** are true and the ***Reason*** is the correct explanation of the ***Assertion*** .

(2) If both ***Assertion*** and ***Reason*** are true but ***Reason*** is not the correct explanation of the ***Assertion*** .

(3) If ***Assertion*** is true but ***Reason*** is false.

(4) If ***Assertion*** is false but ***Reason*** is true.

PO0074

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EXERCISE # O-II

1. In sodium fusion test of organic compounds the nitrogen of the organic compound is converted into

(1) Sodamide (2) Sodium cyanide (3) Sodium nitrite (4) Sodium nitrate

PO0012

2. The Lassaigne's extract is boiled with dil. HNO_3 before testing for halogens because

(1) Silver halides are soluble in HNO_3 (2) Na_2S and NaCN are decomposed by HNO_3

(3) Ag_2S is soluble in HNO_3 (4) AgCN is soluble in HNO_3 **PO0013**

3. In a Lessaigne's test for nitrogen, the blue colour is due to the formation of

(1) Potassium ferricyanide (2) Sodium cyanide

(3) Sodium ferrocyanide (4) Ferri-ferro cyanide **PO0014**

4. Which of the following compounds does not show Lassaigne test for nitrogen?

(1) Urea (2) Hydrazine (3) Phenyl hydrozene (4) Azo benzene

PO0015

5. When piece of human hair are heated srongly with soda lime smell of ammonia can be detected. which one of the following conclusion can be drawn from this observation.

(1) Ammonia is present in human hair (2) Ammonium salt is present in hair

(3) Hair contains amino acids (4) None of the above **PO0016**

6. The Beilstein test for organic compounds is used to detect

(1) Nitrogen (2) Sulphur (3) Carbon (4) Halogen **PO0017**

7. Which of the following is the best scientific method to test presence of water in liquid?

(1) Smell (2) Taste

(3) Use of litmus paper (4) Use of anhydrous copper sulphate **PO0018**

8. Liebig's method is used for the estimation of

(1) Nitrogen (2) Sulphur

(3) Carbon and hydrogen (4) Halogens **PO0019**

9. 0.765 gm of an acid gives 0.535 gm of CO_2 and 0.138 gm of H_2O. Then the ratio of percentage of carbon and hydrogen is

(1) 19:2 (2) 18:11 (3) 20:17 (4) 1:7 **PO0020**

10. Nitrogen is an organic compound can be estimated by

(1) Kjeldahl's method only (2) Duma's method only

(3) Both the methods (4) none of these methods **PO0021**

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11. Dumas method involves the determination of nitrogen content in the organic compound in form of

(1) NH_3 (2) N_2 (3) NaCN (4) $(NH_4)_2SO_4$ **PO0022**

12. In Kjeldahl's method, nitrogen present is estimated as

(1) N_2 (2) NO (3) NH_3 (4) NO_2 **PO0023**

13. The percentage of sulphur in an organic compound whose 0.32g produces 0.233g of $BaSO_4$ [At.wt.Ba=137, S=32] is

(1) 1.0 (2) 10.0 (3) 23.5 (4) 32.1 **PO0024**

14. If 0.2 gram of an organic compound containing carbon, hydrogen and oxygen on combustion, yielded 0.147 gram carbon dioxide and 0.12 gram water. What will be the content of oxygen in the substance?

(1) 73.29% (2) 78.45% (3) 83.23% (4) 89.50% **PO0025**

15. Simple distillation can be used to separate liquids which differ in their boiling points at least by

(1) 5°C (2) 10°C (3) 35 - 50°C (4) 100°C **PO0036**

16. Molecular mass of a volatile substance may be obtained by

(1) Kjeldahl's method (2) Duma's method

(3) Victor meyer's method (4) Liebig's method **PO0047**

17. Kjeldahl's method can not be used for the estimation of Nitrogen in

(1) Pyridine (2) Nitro compounds (3) Azo compounds (4) All **PO0049**

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EXERCISE # S-I

1. 0.2585 g of an organic compound containing iodine was heated with excess of strong nitric aicd and silver nitrate in carius tube. The precipitate of silver iodide was filtered washed and dried. It's weight was formed to be 0.3894 g calculate the % of iodine in the compounds.

PO0075

2. 0.185 g of an organic substance when treated with concentrated HNO_3 gave 0.320 g of silver bromide calculate the % of bromine in the compound.

PO0076

3. During nitrogen estimation of an organic compound by Kjeldahl method the ammonia evolved by 0.5 g of the compound neutralised 10 ml of 1 M H_2SO_4. Calculate % of nitrogen in the compounds.

PO0077

4. 0.35 g of an organic compounds was Kjeldahlised and ammonia obtained was passed into 100 ml of $\frac{M}{10}$ H_2SO_4. The excess acid required 154 ml of $\frac{M}{10}$ NaOH for neutalisation, calculate the % nitrogen in the compounds.

PO0078

5. 0.2475 g of an organic compound gave on combustion 0.4950 g of CO_2 and 0.2025 g of H_2O. If % of C and hydrogen is x and y than x-2y will be :

PO0079

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EXERCISE # JEE-MAINS

1. The compound formed in the positive test for nitrogen with the Lassaigne solution of an organic compound is - **[AIEEE-2004]**

(1) $Fe_4[Fe(CN)_6]_3$ (2) $Na_3[Fe(CN)_6]$

(3) $Fe(CN)_3$ (4) $Na_4[Fe(CN)_5)NOS]$

PO0051

2. The distillation technique most suited for separating glycerol from spent-lye in the soap industry is : **[JEE MAIN 2016]**

(1) Distillation under reduced pressure (2) Simple distillation

(3) Fractional distillation (4) Steam distillation

PO0052

3. Given below are two statements : **[JEE MAIN 2019]**

Statement I : A mixture of chloroform and aniline can be separated by simple distillation.

Statement II : When separating aniline from a mixture of aniline and water by steam distillation aniline boils below its boiling point.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement-I is false but statement II is true

(2) Both statement-I and statement II are false

(3) Statement-I is true but statement II is false

(4) Both statement-I and statement II are true

PO0053

4. The correct match between item I and item II is : **[JEE MAIN 2019]**

	Item-I (mixture)		**Item-II (sepration method)**
(A)	H_2O : sugar	(P)	Sublimation
(B)	H_2O : Aniline	(Q)	Recrystallization
(C)	H_2O : Toluene	(R)	Steam disillation
		(S)	Differential extraction

(1) A $\rightarrow$ S, B $\rightarrow$ R, C $\rightarrow$ P (2) A $\rightarrow$ Q, B $\rightarrow$ R, C $\rightarrow$ P

(3) A $\rightarrow$ R, B $\rightarrow$ P, C $\rightarrow$ S (4) A $\rightarrow$ Q, B $\rightarrow$ R, C $\rightarrow$ S

PO0054

5. If dichloromethane (DCM) and water (H_2O) are used for differential extraction, which one of the following statements is correct? **[JEE MAIN 2019]**

(1) DCM and H_2O would stay as upper and lower layer respectively in the separating funnel (S.F.)

(2) DCM and H_2O would stay as lower and upper layer respectively in the S.F.

(3) DCM and H_2O will make turbid / colloidal mixture.

(4) DCM and H_2O will be miscible clearly

PO0055

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6. The principle of column chromatography is : **[JEE MAIN 2019]**

(1) Gravitational force

(2) Capillary action

(3) Differential adsorption of the substances on the solid phase

(4) Differential absorption of the substances on the solid phase

PO0056

7. In chromatography, which of the following statements is incorrect for R_f?

(1) R_f value depends on the type of chromatography. **[JEE MAIN 2019]**

(2) Higher R_f value means higher adsorption.

(3) R_f value is dependent on the mobile phase.

(4) The value of R_f can not be more than one.

PO0057

8. A chromatography column, packed with silica gel as stationary phase, was used to separate a mixture of compounds consisting of (A) benzanilide (B) aniline and (C) acetophenone. When the column is eluted with a mixture of solvents, hexane : ethyl acetate (20 : 80), the sequence of obtained compounds :

(1) (B), (C) and (A) (2) (C), (A) and (B) **[JEE MAIN 2020]**

(3) (A), (B) and (C) (4) (B), (A) and (C)

PO0058

9. In chromotography technique, the purification of compound is independent of :

(1) Mobility or flow of solvent system **[JEE MAIN 2020]**

(2) Solubility of the compound

(3) Length of the column or TLC Plate

(4) Physical state of the pure compound

PO0059

10. Given below are two statements : **[JEE MAIN 2020]**

Statement-I : Retardation factor (R_f) can be measured in meter/centimeter.

Statement-II : R_f value of a compound remains constant in all solvents.

Choose the most appropriate answer from the options given below:

(1) Statement-I is true but statement-II is false

(2) Both statement-I and statement-II are true

(3) Both statement-I and statement-II are false

(4) Statement-I is false but statement-II is true

PO0060

11. Kjeldahl's method cannot be used to estimate nitrogen for which of the following compounds?

(1) $C_6H_5NO_2$ (2) $C_6H_5NH_2$ **[JEE MAIN 2020]**

(3) $CH_3CH_2–C\equiv N$ (4) $NH_2–\overset{\overset{O}{\|}}{C}–NH_2$

PO0061

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12. Match List-I with List-II **[JEE MAIN 2021]**

List-I Test/Reagents/Observation(s)	List-II Species detected
(a) Lassaigne's Test	(i) Carbon
(b) Cu(II) oxide	(ii) Sulphur
(c) Silver nitrate	(iii) N, S, P, and halogen
(d) The sodium fusion extract gives black precipitate with acetic acid and lead acetate	(iv) Halogen Specifically

The correct match is :

(1) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(2) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

(3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(4) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

PO0062

13. Which one of the following set of elements can be detected using sodium fusion extract ? **[JEE MAIN 2021]**

(1) Sulfur, Nitrogen, Phosphorous, Halogens

(2) Phosphorous, Oxygen, Nitrogen, Halogens

(3) Nitrogen, Phosphorous, Carbon, Sulfur

(4) Halogens, Nitrogen, Oxygen, Sulfur

PO0063

14. In Carius method, halogen containing organic compound is heated with fuming nitric acid in the presence of : **[JEE MAIN 2021]**

(1) HNO_3 (2) $AgNO_3$ (3) $CuSO_4$ (4) $BaSO_4$

PO0064

15. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R) : **[JEE MAIN 2021]**

Assertion (A) : A simple distillation can be used to separate a mixture of propanol and propanone.

Reason (R) : Two liquids with a difference of more than 20°C in their boiling points can be separated by simple distillations.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) (A) is false but (R) is true.

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

PO0065

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16. Using the provided information in the following paper chromatogram : **[JEE MAIN 2020]**

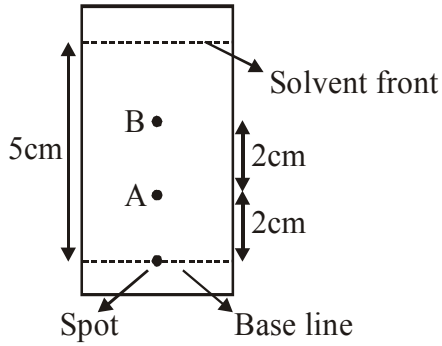


Figure : Paper chromatography for compounds A and B.

the calculate R_f value of A _____ $\times 10^{-1}$.

PO0066

17. 0.8 g of an organic compound was analysed by Kjeldahl's method for the estimation of nitrogen. If the percentage of nitrogen in the compound was found to be 42%, then _____ mL of 1 M H_2SO_4 would have been neutralized by the ammonia evolved during the analysis. **[JEE MAIN 2021]**

PO0067

18. When 0.15 g of an organic compound was analyzed using Carius method for estimation of bromine, 0.2397 g of AgBr was obtained. The percentage of bromine in the organic compound is _____.

(Nearest integer)

[Atomic mass : Silver = 108, Bromine = 80] **[JEE MAIN 2021]**

PO0068

19. The transformation occurring in Duma's method is given below : **[JEE MAIN 2021]**

$$C_2H_7N + \left(2x + \frac{y}{2}\right)CuO \rightarrow xCO_2 + \frac{y}{2}H_2O + \frac{z}{2}N_2 + \left(2x + \frac{y}{2}\right)Cu$$

The value of y is _____. (Integer answer) **PO0069**

20. Phenol on reaction with dilute nitric acid, gives two products. Which method will be most effective for large scale separation ? **[JEE MAIN 2022]**

(1) Chromatographic separation

(2) Fractional Crystallisation

(3) Steam distillation

(4) Sublimation

PO0080

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21. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R. **[JEE MAIN 2022]**

Assertion A: A mixture contains benzoic acid and napthalene. The pure benzoic acid can be separated out by the use of benzene.

Reason R: Benzoic acid is soluble in hot water.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

PO0081

22. During halogen test, sodium fusion extract is boiled with concentrated HNO_3 to **[JEE MAIN 2022]**

(1) remove unreacted sodium

(2) decompose cyanide or sulphide of sodium

(3) extract halogen from organic compound

(4) maintain the pH of extract

PO0082

23. Given below are two statements :

Statement I : In 'Lassaigne's Test, when both nitrogen and sulphur are present in an organic compound, sodium thiocyanate is formed. **[JEE MAIN 2022]**

Statement II : If both nitrogen and sulphur are present in an organic compound, then the excess of sodium used in sodium fusion will decompose the sodium thiocyanate formed to give NaCN and Na_2S.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Statement I and Statement II are correct.

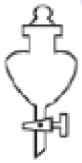
(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

PO0083

24. Which of the following is structure of a separating funnel? **[JEE MAIN 2022]**

(1) 

(2) 

(3) 

(4) 

PO0084

25. Which one of the following techniques is not used to spot components of a mixture separated on thin layer chromatographic plate? **[JEE MAIN 2022]**

(1) I_2 (Solid)

(2) U.V. Light

(3) Visualisation agent as a component of mobile phase

(4) Spraying of an appropriate reagent

PO0085

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26. The formula of the purple colour formed in Laissaigne's test for sulphur using sodium nitroprusside is **[JEE MAIN 2022]**

(1) $NaFe[Fe(CN)_6]$ (2) $Na[Cr(NH_3)_2(NCS)_4]$

(3) $Na_2[Fe(CN)_5(NO)]$ (4) $Na_4[Fe(CN)_5(NOS)]$

PO0086

27. 0.2 g of an organic compound was subjected to estimation of nitrogen by Dumas method in which volume of N_2 evolved (at STP) was found to be 22.400 mL. The percentage of nitrogen in the compound is____.[nearest integer] **[JEE MAIN 2022]**

(Given: Molar mass of N_2 is 28 mol^{-1}. Molar volume of N_2 at STP : 22.4 L)

PO0087

28. 0.25 g of an organic compound containing chlorine gave 0.40 g of silver chloride in Carius estimation. The percentage of chlorine present in the compound is _____. [in nearest integer]

(Given: Molar mass of Ag is 108 g mol^{-1} and that of Cl is 35.5 g mol^{-1}) **[JEE MAIN 2022]**

PO0088

29. In the estimation of bromine, 0.5 g of an organic compound gave 0.40 g of silver bromide. The percentage of bromine in the given compound is _____% (nearest integer)

(Relative atomic masses of Ag and Br are 108u and 80u, respectively). **[JEE MAIN 2022]**

PO0089

30. In Carius method of estimation of halogen. 0.45 g of an organic compound gave 0.36 g of AgBr. Find out the percentage of bromine in the compound. **[JEE MAIN 2022]**

(Molar masses : AgBr = 188 g mol^{-1}: Br = 80 g mol^{-1})

(1) 34.04% (2) 40.04%

(3) 36.03% (4) 38.04%

PO0090

31. Match List-I with List-II **[JEE MAIN 2022]**

List-I (Mixture)	**List-II (Purification Process)**
(A) Chloroform & Aniline	(I) Steam distillation
(B) Benzoic acid & Napthalene	(II) Sublimation
(C) Water & Aniline	(III) Distillation
(D) Napthalene & Sodium chloride	(IV) Crystallisation

(1) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(3) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(4) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

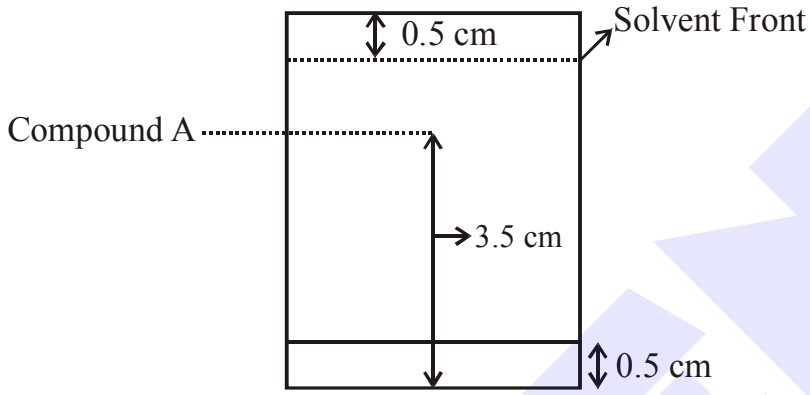
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32. While estimating the nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from 0.25 g of the compound neutralized 2.5 mL of 2 M H_2SO_4. The percentage of nitrogen present in organic compound is . . . . **[JEE MAIN 2022]**

PO0092

33. Following chromatogram was developed by adsorption of compound 'A' on a 6 cm TLC glass plate. Retardation factor of the compound 'A' is _____ $\times 10^{-1}$. **[JEE MAIN 2023]**



PO0093

34. Match List I with List II: **[JEE MAIN 2023]**

	List I (Mixture)		List II (Separation Technique)
(A)	$CHCl_3 + C_6H_5NH_2$	I	Steam distillation
(B)	$C_6H_{14} + C_5H_{12}$	II	Differential extraction
(C)	$C_6H_5NH_2 + H_2O$	III	Distillation
(D)	Organic compound in H_2O	IV	Fractional distillation

(1) A-IV, B-I, C-III, D-II

(2) A-III, B-IV, C-I, D-II

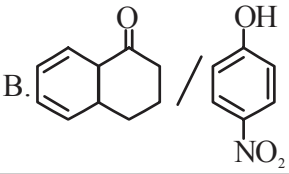
(3) A-II, B-I, C-III, D-IV

(4) A-III, B-I, C-IV, D-II

PO0094

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35. Match items of column I and II **[JEE MAIN 2023]**

Column I (Mixture of compounds)	Column II (Separation Technique)
A. H_2O/CH_2Cl_2	i. Crystallization
B. 	ii. Differential solvent extraction
C. Kerosene/Naphthalene	iii. Column chromatography
D. $C_6H_{12}O_6$/NaCl	iv. Fractional Distillation

Correct match is :

(1) A-(iii), B-(iv), C-(ii), D-(i)
(2) A-(i), B-(iii), C-(ii), D-(iv)
(3) A-(ii), B-(iii), C-(iv), D-(i)
(4) A-(ii), B-(iv), C-(i), D-(iii)

PO0095

36. In Dumas method for the estimation of , the sample is heated with copper oxide and the gas evolved is passed over : **[JEE MAIN 2023]**

(1) Ni (2) Copper gauze (3) Pd (4) Copper oxide

PO0096

37. Compound that will give positive Lassaigne's test for both nitrogen and halogen is **[JEE MAIN 2023]**

(1) $N_2H_4.HCl$
(2) $CH_3NH_2.HCl$
(3) NH_4Cl
(4) $NH_2OH.HCl$

PO0097

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ANSWER-KEY

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	1	2	4	3	3	1	2	4
Que.	11	12	13	14	15					
Ans.	4	1	4	2	2					

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	4	2	3	4	4	3	1	3
Que.	11	12	13	14	15	16	17			
Ans.	2	3	2	1	3	3	4			

EXERCISE # S-I

1. 81.4%

Sol. w = 0.2585 g

wt of AgI = 0.3894 g

$$\% \text{ of I} = \frac{127}{235}\times\frac{0.3894}{0.2585}\times 100 = 81.4\ \%$$

2. 73.60%

Sol. w = 0.185 g

wt of AgBr = 0.320 g

$$\% \text{ of Br} = \frac{80\times 0.320\times 100}{188\times 0.185} = 73.60\%$$

3. 56.06

Sol. 1 M of 10 ml H_2SO_4 = 1 M of 20 ml NH_3

1000 ml of 1 M ammonia contains = 14 g N

$$\therefore \text{ 200 ml of 1 M ammonia contains} = \frac{14\times 20}{1000}\text{g}$$

$$\therefore \ \% \text{ of N} = \frac{14\times 20}{1000\times 0.5}\times 100 = 56.0\%$$

4. 18.40

Sol. Volume of $\frac{M}{10}$ H_2SO_4 taken = 100 ml

excess volume of $\frac{M}{10}$ H_2SO_4 is

154 ml of $\frac{M}{10}$ NaOH = $\frac{154}{2}$ ml of $\frac{M}{10}$ H_2SO_4

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$\therefore$ volume of $\frac{M}{10}$ H_2SO_4 left unused = 77 ml

Volume of $\frac{M}{10}$ H_2SO_4 used for neutralisation of NH_3 = 100 – 77 – 23 ml

Now, 23 ml of $\frac{M}{10}$ H_2SO_4 = 2 × 23 ml of $\frac{M}{10}$ NH_3

= 46 ml of $\frac{M}{10}$ NH_3 now, 1000 ml of 1 M NH_3 contains 14 g N

$\therefore$ 46 ml of $\frac{M}{10}$ NH_3 contain $\frac{14}{1000}\times\frac{46\times1}{10}$

% of N = $\frac{14\times46\times100}{1000\times10\times0.35}=18.4\%$

5. 36.36

Sol. wt of organic compounds

(w) = 0.247 5g

wt of CO_2(b) = 0.4950

wt of H_2O (a) = 0.2025 g

% of C = $\frac{12}{44}\times\frac{b}{w}\times100$

= $\frac{12}{44}\times\frac{0.4950}{0.2475}\times100$ = 54.54

% of H = $\frac{2}{18}\times\frac{a}{w}\times100$

= $\frac{2}{18}\times\frac{0.2025}{0.2475}\times100$ = 9.09

Now x = 54.54

y = 9.04

x – 2y = 54.54 – 2 × 9.09

= 54.54 – 18.18 = 36.36

EXERCISE # JEE-MAINS

1. (1)	**2. (1)**	**3. (4)**	**4. (1)**	**5. (2)**
6. (3)	**7. (2)**	**8. (2)**	**9. (4)**	**10. (3)**
11. (1)				

Sol. Kjeldahl's method for estimation of nitrogen is not applicable for nitrobenzene $C_6H_5NO_2$. because reaction with H_2SO_4, nitrobenzene can not give ammonia.

12. (3)	**13. (1)**	**14. (2)**	**15. (4)**	**16. (4)**
17. (12)	**18. (68)**	**19. (7)**	**20. (3)**	

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21.(4)

Sol. Benzoic acid and Napthalene can be effectively separated by crystallization. Benzoic acid is soluble in hot water whereas Napthalene is insoluble.

Hence assertion is incorrect but reason is correct

22. (2)

Sol. Sodium fusion extract is boiled with concentrated HNO3 to remove sodium cyanide and sodium sulphide

23. (1)

Sol. Both statement I & statement II are correct.

24. (1)

Sol. It is used to separate liquid-liquid mixture which is immiscible with different densities

25. (3)

26. (4)

Sol. $Na_2S + Na_2[Fe(CN)_5NO] \rightarrow Na_4[Fe(CN)_5NOS]$

27. (14)

Sol. weight of organic compound = 0.2g

mass of $N_2(g)$ evolved = $\frac{22.4\times10^{-3}}{22.4}\times28 = 28 \times 10^{-3}$g

$\%$ of $N = \frac{28\times10^{-3}}{0.2}\times100 = 14$

28. (40)

Sol. wt. of organic compound = 0.25 g

mass of $Cl = \frac{35.5}{143.5}\times0.4$g

mass % of Cl in the organic compound

$= \frac{35.5\times0.4}{143.5\times0.25}\times100 = 39.58\%$

29. (34)

Sol $\underset{0.5\,g}{O.C} \longrightarrow \underset{0.4\,g}{AgBr}$

mol of Br = mol of AgBr = $\frac{0.4}{188}$

$\%$ Br = $\%Br = \frac{\frac{0.4}{188}\times80}{0.5}\times100 = 34.04\%$

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30. (1)

Sol. Mass of organic compound = 0.45 gm

Mass of AgBr obtained = 0.36 gm

$$\text{Moles of AgBr} = \frac{0.36}{188}$$

$$\text{Mass of Bromine} = \frac{0.36}{188} \times 80 = 0.1532\text{gm}$$

$$\%\ \text{Br in compound} = \frac{0.1532}{0.45} \times 100 = 34.04\%$$

31. (4)

Sol. (1) Chloroform + Aniline → (III) Distillation

(2) Benzoic acid + Napthalene → (IV) Crystallisation

(3) Water + Aniline → (I) Steam distillation

(4) Napthalene + Sodium chloride → (II) Sublimation

32. (56)

Sol. $$\%N = \frac{1.4(N_1V_1)}{\text{mass of organic compound}}$$

$$\%N = \frac{1.4(2.5 \times 2 \times 2)}{0.25} = 56$$

33. (6)

Sol. $$R_f = \frac{\text{Distance moved by the substance from base line}}{\text{Distance moved by the solvent from base line}}$$

$$= \frac{3.0\text{ cm}}{5.0\text{ cm}} = 0.6 \text{ or } 6 \times 10^{-1}$$

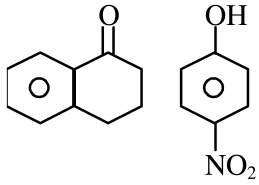
34. (2)

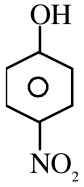
List I (Mixture)	**List II (Separation Technique)**
$CHCl_3 + C_6H_5NH_2$	Distillation
$C_6H_{14} + C_5H_{12}$	Fractional distillation
$C_6H_5NH_2 + H_2O$	Steam distillation
Organic compound in H_2O	Differential extraction

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35. (3)

Sol. **A.** H_2O/CH_2Cl_2 ii, $CH_2Cl_2 > H_2O$ (density) so they can be separated by differential solvent extraction.

B.  iii. column chromatography

Due to H-bonding in  it can be separated from by column chromatography.

C. Due to different B.P. of kerosene and Naphthalene it can be separated by fractional distillation.

D. $C_6H_{12}O_6$/NaCl i. Crystallization.

NaCl (ionic compound) can be crystallized.

36. (2)

Sol. Duma's method.

The nitrogen containing organic compound, when heated with CuO in a atmosphere of CO_2, yields free N_2 in addition to CO_2 and H_2O.

$$C_xH_yN_z + (2x + \frac{y}{2})\,CuO \rightarrow$$

$$x\,CO_2 + \frac{y}{2}H_2O + \frac{z}{2}N_2 + (2x + \frac{y}{2})\,Cu$$

Traces of nitrogen oxides formed, if any, are reduced to nitrogen by passing the gaseous mixture over heated copper gauze.

37. (2)

Sol. $CH_3NH_2 \,.\, HCl \xrightarrow[\text{fusion}]{\text{Na}}$ NaCN and NaCl

NaCN gives +ve test for nitrogen and

NaCl gives +ve test for halogen

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Important Notes

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